Exercises

Ruben Lliuyacc Blas

June 12, 2026

1 Problem 1

Draw the single-phase equivalent circuit of the power system shown in Figure 1. Use the exact models for the transformers and the transmission line. For this exercise, you can model the loads as purely resistive impedances.

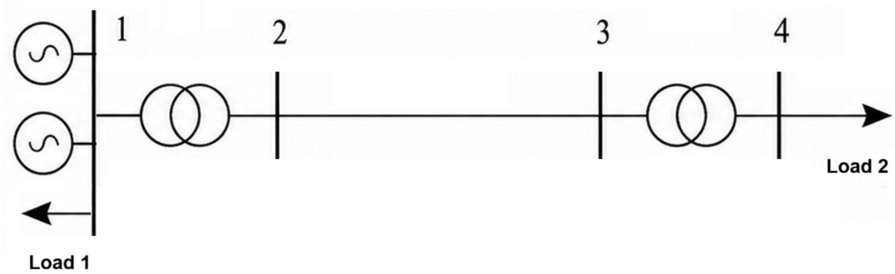

Figure 1: Unifilar diagram of the network.

2 Problem 2

A step-up transformer used in a hydroelectric generating station has a rated power of 100 MVA and a rated voltage of 13.8 kV/220 kV. The results of the standard transformer tests are:

- **Open-circuit test**

$$I_0 = 0.8\%, \qquad P_0 = 100\,\mathrm{kW}$$

- **Short-circuit test**

$$V_{sc} = 8\%, \qquad P_{sc} = 300\,\mathrm{kW}$$

Determine the per-unit equivalent circuit of the transformer.

2.1 Answer

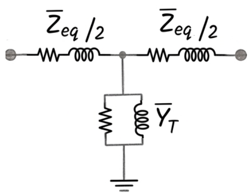

Figure 2: Per-unit equivalent circuit of the transformer.

$$Z_{eq} = 0.003 + j0.07994\,\Omega, \qquad Y_t = 0.001 - j0.00794\,\Omega^{-1}$$

3 Problem 3

Obtain the per-unit diagram of the circuit in Figure 3 using, on the transmission lines, a base power and a base voltage of 100 MVA and 220 kV, respectively.

The characteristics and rated values for each network element are given in Table 1.

It is also known that at bus *4* a load of 70 Mvar and 0 MW is consumed, and at bus *6* a load of 0 Mvar and 60 MW is consumed. In addition, if the voltage at bus *2* is 231 kV, calculate the voltages and currents at buses *4* and *6*.

Table 1: Network data

Element	$V_{\mathbf{nom}}$ **(kV)**	$S_{\mathbf{nom}}$ **(MVA)**	**Impedance**
Generator	24	200	$X_G = 10\%$
T1	25/230	200	$X_{cc} = 9\%$
T2	220/130	100	$X_{cc} = 10\%$
T3	240/66	75	$X_{cc} = 8\%$
Line 23	—	—	$Z = 10 + j60\ \Omega$
Line 25	—	—	$Z = 50j\ \Omega$

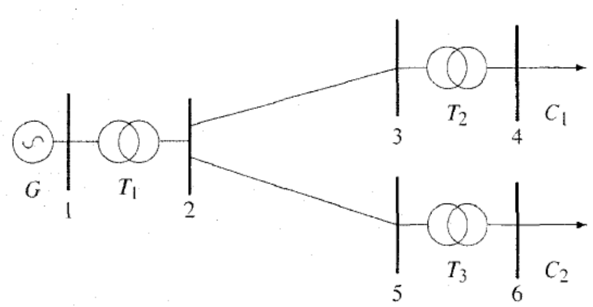

Figure 3: Single-line diagram of the network

3.1 Answer

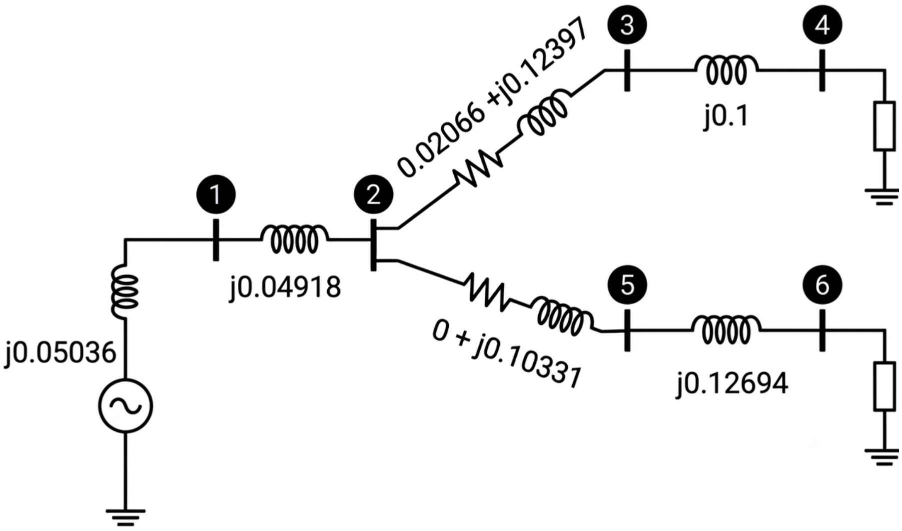

Figure 4: Per-unit equivalent circuit of the transformer.

Having the reference angle $\angle 0^\circ$ at bus 6.

$$V_4 = 0.86957\,\angle 8.17^\circ \text{ p.u.} = 113.045\,\angle 8.17^\circ \text{ kV}, \qquad I_4 = 0.80499\,\angle{-81.83^\circ} \text{ p.u.} = 357.51\,\angle{-81.83^\circ} \text{ A}$$

$$V_6 = 1.04158\,\angle 0^\circ \text{ p.u.} = 63.016\,\angle 0^\circ \text{ kV}, \qquad I_6 = 0.57629\,\angle 0^\circ \text{ p.u.} = 549.95\,\angle 0^\circ \text{ A}$$